

OS & Computer Networks (ETE2AE)

Active Recall Memorization Checklist & Quick Cram Study Sheet

Rapid self-testing tool. Attempt to answer each question before reading the 'Key Concepts' checklist. Check off your status.

OS Basics & Architecture

Master ed	Review	Question & Core Concepts Checklist
[]	[]	OS Purpose & Resources <i>What is the primary purpose of an Operating System, and what are the three key system resources it controls and allocates?</i> Recall Keys: The Operating System controls and assigns system resources: CPU time, operating memory (RAM), and local storage space (HDD/SSD). It acts as a bridge between the user/applications and the hardware, and provides APIs for user programs.
[]	[]	5 Elements of a Modern OS <i>What are the 5 core components/elements that compose a modern operating system?</i> Recall Keys: 1. Kernel: The core program that links applications to the hardware. 2. File System: Responsible for organizing and storing data on local storage devices (LTSDs). 3. Programming Language: The languages used to write the OS (e.g., C, Assembly). 4. User Interface (UI): The mechanism for human interaction, either graphical (GUI) or text-based (CLI/Shell). 5. API (Application Programming Interface): Set of functions allowing applications to request services from the kernel.
[]	[]	Cooperative vs. Preemptive Multitasking <i>What is the key operational difference between Cooperative and Preemptive multitasking? Name the systems that used/use them.</i> Recall Keys: • Cooperative Multitasking: Applications voluntarily yield CPU control back to the OS. If a program crashes or hangs, the entire system freezes (used in early Windows 3.x, Classic Mac OS). • Preemptive Multitasking: The OS uses timer interrupts to forcibly reclaim the CPU from running applications. A hung application can be terminated without bringing down the system (used in modern OSs: Windows NT/95+, macOS X+, Linux).
[]	[]	Desktop vs. Mobile OS Design <i>How do mobile operating systems differ in design requirements and constraints compared to desktop systems?</i> Recall Keys: Mobile OSs are designed for battery-operated devices, utilize a touchscreen as the primary interface, expect frequent display orientation changes, restrict background processing to conserve power, and implement stricter security/sandboxing due to a high physical risk of theft/loss.
[]	[]	Type 1 vs. Type 2 Hypervisors <i>Contrast Type 1 (bare-metal) and Type 2 (hosted) hypervisors. Provide examples of each.</i> Recall Keys: • Type 1 (Bare-Metal): Runs directly on the physical hardware. Highly efficient, secure, and used in datacenters (e.g., VMware ESXi, Proxmox VE, Microsoft Hyper-V). • Type 2 (Hosted): Runs as an application on top of an existing host Operating System. Has more overhead (e.g., Oracle VirtualBox, VMware Workstation).
[]	[]	Open Source Licenses: Copyleft vs. Permissive <i>Explain the difference between Copyleft and Permissive open-source licenses. Give examples.</i> Recall Keys: • Copyleft (e.g., GPL - GNU General Public License): Requires that any modified or derivative software must also be released as open-source under the exact same license terms. • Permissive (e.g., MIT, Apache): Allows users to modify, redistribute, and even close the source code for commercial use with minimal restrictions (only attribution is required).

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[]	[]	BIOS Location and Role <i>What is the function of the BIOS software, and where is it physically stored on the computer?</i> Recall Keys: The BIOS (Basic Input/Output System) initializes and tests system hardware components during the boot process (POST - Power-On Self-Test) and loads the OS bootloader. It is physically stored in non-volatile EEPROM or NAND flash memory on the motherboard.
[]	[]	Linux Kernel Definition & History <i>What is the Linux kernel, and was it derived from Unix code?</i> Recall Keys: Linux is a free, open-source monolithic kernel written completely from scratch by Linus Torvalds in 1991. It is Unix-like but not derived from original Unix code. It handles process scheduling, memory allocation, and hardware communication.
[]	[]	Cloud Computing & 5 Characteristics <i>Define cloud computing and list the five key characteristics of a cloud service as defined in ICT.</i> Recall Keys: Cloud computing is the delivery of computing services (servers, storage, databases, networking, software) over the Internet. The five key characteristics are: 1. On-Demand Self-Service: Users can provision resources automatically without human interaction with the provider. 2. Broad Network Access: Services are accessible over the network from anywhere using various devices (phones, laptops). 3. Location-Independent Resource Pooling: Provider's resources are pooled to serve multiple consumers in a multi-tenant model. 4. Rapid Elasticity: Resources can be rapidly provisioned and released to scale out or in based on demand. 5. Measured Service: Resource usage is monitored, controlled, and optimized dynamically, allowing pay-per-use billing.

Process Management

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[]	[]	5 Process States & Transitions <i>Name the 5 states of a process lifecycle and explain what triggers the transitions between them.</i> Recall Keys: 1. New: The process is being created. 2. Ready: The process is waiting in queue to be assigned to a CPU. 3. Running: Instructions are being executed by the CPU. 4. Waiting/Blocked: The process is waiting for an event or I/O operation to complete. 5. Terminated: The process has finished execution. *Transitions*: Ready → Running (OS Scheduler dispatch); Running → Ready (time slice interrupt); Running → Waiting (I/O request); Waiting → Ready (I/O complete).
[]	[]	Process vs. Thread <i>Compare a process and a thread in terms of resources, memory, PID, and context-switching cost.</i> Recall Keys: • Process: An independent program in execution. Has its own virtual address space, memory, open files, and a unique Process ID (PID). Context switching is expensive. • Thread: A lightweight execution unit within a process. It shares the parent process's memory space, PID, and resources, but has its own stack and registers. Context switching is cheap.
[]	[]	First-Come, First-Served (FCFS) Scheduling <i>Explain FCFS CPU scheduling, its preemptive status, and define the 'Convoy Effect'.</i> Recall Keys: FCFS is a non-preemptive algorithm where the CPU is allocated to processes in their arrival order. The Convoy Effect occurs when short processes are stuck waiting behind a single, long CPU-bound process, resulting in high average waiting times.
[]	[]	Shortest Job First (SJF) Scheduling <i>Explain SJF CPU scheduling, its optimality, its starvation risk, and its primary implementation challenge.</i> Recall Keys: SJF selects the process with the shortest next CPU burst. It is non-preemptive and yields the mathematically optimal average waiting time . However, it is impossible to implement in general-purpose OSs because the future CPU burst duration cannot be predicted. It can also cause starvation of long processes if short processes keep arriving.
[]	[]	Round Robin (RR) Scheduling <i>Explain Round Robin scheduling and how the choice of time quantum (Q) affects system performance.</i> Recall Keys: Round Robin is a preemptive algorithm designed for time-sharing systems. Each process gets a small fixed slice of CPU time (time quantum Q). If unfinished, it is preempted and sent to the back of the queue. • Q is too small: Causes high context-switch overhead, wasting CPU cycles. • Q is too large: Behaves exactly like FCFS.
[]	[]	4 Deadlock Conditions <i>Name the 4 conditions that must hold simultaneously for a deadlock to occur.</i> Recall Keys: 1. Mutual Exclusion: At least one resource must be held in a non-shareable mode (only one process can use it at a time). 2. Hold and Wait: A process must hold at least one resource and wait to acquire additional resources held by other processes. 3. No Preemption: Resources cannot be forcibly taken from a process; they can only be released voluntarily. 4. Circular Wait: A closed loop of processes exists where each process holds a resource that the next process in the loop is waiting to acquire.

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☐	☐	Banker's Algorithm <i>What is the Banker's Algorithm, what deadlock strategy does it represent, and how does it determine allocation safety?</i> Recall Keys: The Banker's Algorithm is a deadlock avoidance algorithm. When a process requests resources, the OS simulates the allocation and checks if it leaves the system in a safe state (where a sequence exists allowing all processes to finish without deadlock). If unsafe, the request is denied or delayed.
☐	☐	Deadlock Prevention vs. Recovery <i>Contrast deadlock prevention and deadlock detection & recovery.</i> Recall Keys: • Deadlock Prevention: Structural methods that ensure at least one of the 4 deadlock conditions cannot hold (e.g., requesting all resources at once to prevent Hold-and-wait, or forcing resource preemption). • Detection & Recovery: Allows deadlocks to occur, runs diagnostic algorithms to detect them, and recovers by either terminating deadlocked processes or preempting resources.

Memory Management

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[]	[]	Virtual Memory & Paging Principle <i>Describe the function and principles of Virtual Memory. How does the CPU access data stored on disk?</i> Recall Keys: Virtual memory compensates for physical RAM shortage by utilizing disk space (e.g., pagefile/swap). The OS divides memory into fixed-size pages (virtual) mapped to physical RAM frames using a Page Table . When the CPU accesses virtual addresses, the MMU (Memory Management Unit) translates them. If the data is on disk, a page fault is triggered, and the OS performs swapping/paging to load it into RAM.
[]	[]	Swapping vs. Paging <i>What is the difference between Swapping and Paging? What is the recommended swap-to-RAM ratio?</i> Recall Keys: • Swapping: Moving an entire process out of physical memory to secondary storage (swap space). • Paging: Moving fixed-size blocks of memory (pages/pagefiles) between RAM and disk. The recommended swap-to-RAM ratio is 3:1 to avoid thrashing while maintaining stability.
[]	[]	Translation Lookaside Buffer (TLB) <i>What is the Translation Lookaside Buffer (TLB) and why is it crucial for virtual memory performance?</i> Recall Keys: The TLB is a high-speed, hardware-implemented cache located inside the CPU's MMU. It stores recent virtual-to-physical address translations. Without the TLB, every memory access would require two lookups (one for the page table in RAM, and one for the actual data), cutting memory performance in half.
[]	[]	FIFO Page Replacement & Belady's Anomaly <i>Explain FIFO page replacement and define Belady's Anomaly.</i> Recall Keys: • FIFO (First-In, First-Out): Replaces the page that has been in memory the longest. • Belady's Anomaly: A phenomenon where allocating more physical page frames to a process results in more page faults. FIFO suffers from this anomaly, whereas stack-based algorithms like LRU do not.
[]	[]	Least Recently Used (LRU) Page Replacement <i>Explain LRU page replacement and why it does not suffer from Belady's Anomaly.</i> Recall Keys: • LRU (Least Recently Used): Replaces the page that has not been accessed for the longest period of time. It is a 'stack algorithm', meaning the set of pages in memory for n frames is always a subset of the pages in memory for $n+1$ frames. Therefore, allocating more frames is guaranteed not to increase page faults.
[]	[]	Optimal Page Replacement Algorithm <i>What is the Optimal Page Replacement algorithm, and why is it impossible to implement in practice?</i> Recall Keys: The Optimal page replacement algorithm replaces the page that will not be used for the longest period of time in the future . It yields the absolute lowest page fault rate. It is impossible to implement in practice because it requires perfect future knowledge of memory access patterns. It is used as a theoretical baseline for comparison.
[]	[]	Internal vs. External Fragmentation <i>Contrast internal and external memory fragmentation, and how paging/segmentation handles them.</i> Recall Keys: • Internal Fragmentation: Wasted space inside an allocated block of memory (e.g., a process is allocated a 4 KB page but only uses 1 KB). Paging suffers from internal fragmentation. • External Fragmentation: Wasted space outside allocated blocks because free memory is broken into small, non-contiguous pieces. Solved by compaction or using paging (which maps non-contiguous frames into a contiguous virtual space).

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[]	[]	Thrashing & Working Set Model <i>Define 'Thrashing' and explain how the Working Set Model prevents it.</i> Recall Keys: • Thrashing: A state where the OS spends more time swapping pages in and out of disk than executing instructions. Occurs when physical RAM is overcommitted. • Working Set Model: Tracks the set of pages actively used by a process in a recent time window. The OS will only run a process if its entire working set can fit in physical RAM, preventing thrashing.
[]	[]	Linux Memory Stats & buff/cache <i>What commands are used to check memory utilization in Linux? Explain why the 'free' RAM often appears extremely low, and explain the 'empty RAM is wasted RAM' philosophy (OS Exercise 2).</i> Recall Keys: • Commands: `free -m`, `top`, or viewing `/proc/meminfo`. • Philosophy: 'empty RAM is wasted RAM'. Linux uses almost all spare RAM as buff/cache to speed up filesystem and disk reads. If a program needs memory, the kernel instantly drops caches and reallocates RAM to the program. Therefore, actual Available memory remains high even if strictly Free RAM is tiny.

File Systems & Sharing Models

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[]	[]	FAT16 Volume Size Limit <i>What is the mathematical reason that the largest volume size in the FAT16 file system is 4 GB?</i> Recall Keys: FAT16 uses 16-bit addressing for clusters, allowing a maximum of $2^{16} = 65,536$ clusters. The maximum cluster size supported is 64 KB. Therefore, the maximum volume size is: $65,536 \text{ clusters} \times 64 \text{ KB/cluster} = 4,194,304 \text{ KB} = 4,096 \text{ MB} = 4 \text{ GB}$.
[]	[]	Linux Inodes Structure <i>What is an inode in Linux filesystems (ext4) and what information does it store? Does it store the filename?</i> Recall Keys: An inode (index node) is a data structure on disk representing a file system object. It stores metadata: file size, owner ID, group ID, access permissions (chmod), timestamps (creation/access/modification), and pointers to disk blocks containing the file content. It does NOT store the filename (which is stored in the directory entry).
[]	[]	Host-Terminal Model <i>Explain the Host-Terminal computational model and its impact on network traffic. What does the network transport?</i> Recall Keys: An early computing model where a central, powerful mainframe (host) holds all data and processing power. Users access it via dumb terminals (keyboard + monitor only). The network only transports raw inputs (keystrokes) and raw outputs (screen redraw commands) — no application logic runs locally.
[]	[]	Batch Processing Model <i>Describe the Batch Processing model, its primary feature, and its advantages/disadvantages.</i> Recall Keys: A computational model where jobs are grouped into batches and processed sequentially by a mainframe without user intervention. The primary feature is offline processing. • Advantages: Repetitive tasks run quickly, optimal CPU utilization, doesn't require user attention during runs. • Disadvantages: Hard to debug, high turnaround times, errors halt the queue, and well-trained personnel are required to operate the computer.
[]	[]	Client-Server & 3-Tier Model <i>Explain the Client-Server architecture and identify the 'Universal Client' in a 3-tier web application model.</i> Recall Keys: • Client-Server: Distributed application structure where clients request resources and servers provide them. • 3-Tier Model: Divides application into Presentation Tier, Logic/Application Tier, and Data Tier. • Universal Client: The WWW browser (web browser), which renders the interface locally while data and logic remain on servers.
[]	[]	Defragmentation: Kernel vs. Utility <i>Explain why disk defragmentation is NOT considered a core function of the OS kernel.</i> Recall Keys: Defragmentation is a user-space utility software, not a kernel routine. The kernel's core responsibilities are process scheduling, memory allocation, and hardware device drivers. Defragmentation reorganizes data blocks at the file system level to make files contiguous, improving read speeds.
[]	[]	Server Definition & Types <i>What is a server in network communication, and what are some common types of servers?</i> Recall Keys: A server is a specialized computer program (or the physical machine running it) that offers services to other computer programs (clients) and their users. Common types include: mail servers, proxy servers, application servers, virtual servers, blade servers, policy servers, and file servers.

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☐	☐	Malicious Batch Agents (Botnets) <i>What is the name of unwanted or malicious software that operates within the autonomous agent batch model?</i> Recall Keys: A Botnet. This is a network of compromised systems running malicious software (bots) as autonomous agents under a central command-and-control (C2) server. It executes coordinated tasks like DDoS attacks, data theft, or spam without direct human interaction.

Computer Networks

Master ed	Review	Question & Core Concepts Checklist
[]	[]	OSI 7-Layer Model & PDUs List the 7 layers of the OSI model in order and state the PDU (Protocol Data Unit) for layers 1 to 5. Recall Keys: 1. Physical: Bits (cables, repeaters, hubs) 2. Data Link: Frames (MAC address, switches) 3. Network: Packets (IP address, routers) 4. Transport: Segments (TCP) / Datagrams (UDP) (ports) 5. Session: Data (sockets, connection control) 6. Presentation: Data (encryption, translation) 7. Application: Data (HTTP, DNS, DHCP, SSH)
[]	[]	TCP vs. UDP Compare TCP and UDP in detail (handshake, reliability, overhead, ordering, use cases). Recall Keys: • TCP (Transmission Control Protocol): Connection-oriented (uses a 3-way handshake SYN → SYN-ACK → ACK to open, and a 4-way handshake to close), reliable (acknowledgments, retransmissions), ordered delivery, flow control, high overhead. (e.g., HTTP, SSH, FTP, SMTP). • UDP (User Datagram Protocol): Connectionless, unreliable (best-effort, no ACKs), unordered, low overhead/latency. (e.g., DNS, VoIP, streaming, CoAP).
[]	[]	DHCP DORA Process Explain the DORA process in DHCP and list the configuration settings it assigns. Recall Keys: 1. Discover: Client broadcasts to find a DHCP server. 2. Offer: DHCP server offers an IP address. 3. Request: Client requests the offered IP. 4. Acknowledge: Server confirms the lease and parameters. *Parameters assigned*: Client IP address, Subnet Mask, Default Gateway (Router IP), and DNS Server IP.
[]	[]	DNS Structure and Operation Explain the distributed nature of DNS and the direction from which DNS resolvers read domain names. Recall Keys: DNS (Domain Name System) is a distributed database that maps domain names to IP addresses. DNS resolvers read domain names from the end (right to left), beginning at the root domain (.), then Top-Level Domain (TLD like .cz), then Second-Level Domain (like czu.cz - Czech University of Life Sciences), down to subdomains.
[]	[]	Private IP Addresses & NAT What are private IP addresses, why were they introduced, and how do devices using them access the public internet? Recall Keys: Private IP addresses (RFC 1918) are reserved for local internal networks (e.g., 192.168.x.x, 10.x.x.x). They were introduced to prevent IPv4 address scarcity. They are non-routable on the public internet. To access the internet, a NAT (Network Address Translation) device (router) translates the private IP to a single public IP.
[]	[]	VLAN Access vs. Trunk Ports Compare VLAN Access Ports and Trunk Ports in 802.1Q switch configurations. Recall Keys: • Access Port: Belongs to a single VLAN. Connects end-user devices (PCs, printers). Frames leaving this port are sent without VLAN tags (untagged). • Trunk Port: Configured to carry traffic for multiple VLANs simultaneously over a single physical link (usually between switches/routers). Uses 802.1Q tagging to keep track of VLAN membership.

Master ed	Review	Question & Core Concepts Checklist
[]	[]	IoT Protocols: MQTT vs. CoAP <i>Contrast the MQTT and CoAP messaging protocols used in IoT.</i> Recall Keys: • MQTT (Message Queuing Telemetry Transport): Publish-Subscribe architecture, runs over TCP (reliable), uses a central broker, ideal for continuous low-bandwidth links. • CoAP (Constrained Application Protocol): Request-Response architecture, runs over UDP (lightweight/RESTful), uses GET/POST/PUT methods, ideal for highly constrained, sleeping nodes.
[]	[]	Wireless Networks: Advantages, Disadvantages, & Bands <i>List the pros and cons of wireless networks, define unlicensed bands, and explain frequency trade-offs.</i> Recall Keys: • Advantages: Mobility, cost-effectiveness (no cabling), quick installation/adaptability. • Disadvantages: Unreliability (interference), lower bandwidth, security risks (e.g., packet sniffing). • Unlicensed Bands: Shared frequency bands (2.4 GHz, 5 GHz) that don't require licensing, meaning they are prone to congestion. • Frequency Trade-off: Higher frequencies (5 GHz) provide more bandwidth but have shorter range and poorer obstacle penetration than lower frequencies (2.4 GHz).
[]	[]	Switch MAC Learning Behavior <i>How does an Ethernet switch learn MAC addresses? Explain the MAC table state before and after network traffic (Packet Tracer 2).</i> Recall Keys: • MAC Learning: Switches perform source-MAC address learning. When an Ethernet frame enters a switch port, the switch reads the source MAC and adds a dynamic mapping to its MAC table. Switch does NOT learn destination MACs this way. • MAC Table State: Before traffic, the table is empty or has only default ports. After a ping or frame is sent, the switch populates the table dynamically. Ports receiving no traffic remain unpopulated. Command: `show mac-address-table`.
[]	[]	ARP Resolution in Ping <i>Explain what happens when PC1 pings PC2 for the first time. Focus on the ARP broadcast and subsequent unicast frames (Packet Tracer 2).</i> Recall Keys: • First Packet (ARP): Since PC1 does not know PC2's MAC, it sends an ARP Broadcast (destination MAC `ff:ff:ff:ff:ff:ff`). The switch floods this broadcast out all ports (except ingress). • Second Packet (ICMP): PC2 responds with a unicast ARP reply. Once MACs are known, subsequent ICMP Echo Request and Reply frames are sent as unicast directly between ports. The switch forwards them without flooding.
[]	[]	IP Subnet Overlap Errors <i>What is an IP subnet overlap/conflict error, how is it diagnosed, and how do you resolve it (Packet Tracer 2)?</i> Recall Keys: • Overlap Error: Occurs when two separate interfaces/subnets are assigned overlapping IP address spaces (e.g., configuring both Network A and Network C with `192.168.1.0/24`). It causes routing errors and address conflicts. • Diagnosis/Resolution: Diagnosed via router configuration errors or packet loss. Resolved by assigning a unique subnet range to one of the conflicting networks (e.g., re-addressing Network C to `192.168.3.0/24`) and updating host IPs and gateways.
[]	[]	IP Node Configuration Parameters <i>What four core IP parameters must be assigned to configure a network interface on a node, and what is the role of each?</i> Recall Keys: 1. IP Address: Uniquely identifies the node on the network. 2. Subnet Mask: Determines the size of the local network and distinguishes local hosts from remote subnets. 3. Default Gateway: The IP address of the local router, needed to forward packets outside the local network. 4. DNS Server: The IP address of a domain name resolver, needed to translate human-readable domain names into numeric IP addresses.

Mastered	Review	Question & Core Concepts Checklist
☐	☐	Unlicensed Frequency Bands & Wireless <i>What does it mean for certain frequency bands (e.g., 2.4 GHz, 5 GHz) to be unlicensed? Why do we use higher frequencies?</i> Recall Keys: • Unlicensed bands: These frequencies are shared and open to the public; anyone can transmit on them without a license, making them prone to interference. • Higher Frequencies (e.g. 5 GHz): We use higher frequencies mainly because of the more bandwidth available there, even though they suffer from shorter range and poorer penetration.

CLI & Diagnostics

Master ed	Review	Question & Core Concepts Checklist
[]	[]	CLI Advantages & Disadvantages <i>What are the primary technical advantages and disadvantages of a Command Line Interface (CLI) compared to a GUI?</i> Recall Keys: • Advantages: Remote access requires very low bandwidth (text-only), consumes very little RAM and CPU, runs on low-resolution displays, and is highly scriptable/automatable. • Disadvantages: Steep learning curve, requires remembering exact commands/syntax, no auto-completion by default in basic shells, typos can be catastrophic, and causes eye strain.
[]	[]	PING Tool <i>What is the function of the PING diagnostic tool, what protocol does it use, and is it available on both Windows and Linux?</i> Recall Keys: PING verifies network-level connectivity to a destination IP. It uses the ICMP (Internet Control Message Protocol) Echo Request and Echo Reply packets (operating at Layer 3, with no port numbers). It is installed by default in both Windows and GNU/Linux.
[]	[]	Traceroute (tracert) <i>How does traceroute function to map a network path? Explain TTL and the differences between Windows and Linux implementations.</i> Recall Keys: Traceroute sends packets with increasing TTL (Time to Live) values starting at 1. Each router decrements TTL by 1. When TTL reaches 0, the router discards the packet and sends an ICMP Time Exceeded message back, revealing its IP. • Windows: Uses <code>tracert</code> which sends ICMP Echo Requests. • Linux: Uses <code>traceroute</code> which sends UDP packets (or ICMP/TCP depending on flags).
[]	[]	Linux Process Management Commands <i>Name the primary Linux commands to view static processes, monitor real-time CPU usage, and terminate a process.</i> Recall Keys: • <code>ps</code>: Displays a static snapshot of current running processes (e.g., <code>ps aux</code>). • <code>top</code> / <code>htop</code>: Provides an interactive, dynamic, real-time view of system processes, CPU, and RAM usage. • <code>kill</code>: Terminates a process by sending a signal (default SIGTERM 15, or SIGKILL 9) to its Process ID (PID) (e.g., <code>kill -9 1234</code>).
[]	[]	Linux Disk & Directory Commands <i>Name the Linux commands to list directory contents, check available disk space, and check directory storage usage.</i> Recall Keys: • <code>ls</code>: Lists directory contents (use <code>ls -la</code> to see hidden files and details). • <code>df</code>: Displays the amount of free and used disk space on mounted file systems (use <code>df -h</code> for human-readable format). • <code>du</code>: Estimates file space usage of a directory recursively (use <code>du -sh *</code> to summarize sizes of subfolders).
[]	[]	Linux File Permissions (chmod) <i>Explain the numeric values used in <code>chmod</code> to set file permissions. What does <code>chmod 755</code> represent?</i> Recall Keys: Permissions are defined for Owner (u), Group (g), and Others (o). Values: Read (r) = 4, Write (w) = 2, Execute (x) = 1. The three digits represent: Owner Group Others. • <code>chmod 755</code>: Owner has 7 ($4+2+1 = rwx$), Group has 5 ($4+1 = r-x$), Others have 5 ($4+1 = r-x$). Often used for directories and executable scripts.

Master ed	Review	Question & Core Concepts Checklist
[]	[]	Linux Diagnostics: w and uname <i>Explain the diagnostic purposes and outputs of the Linux commands 'w' and 'uname -a' (OS Exercise 1).</i> Recall Keys: • w: Shows system uptime, load averages (over 1, 5, and 15 mins), and details of all logged-in users (username, TTY, source IP address, login time, idle duration, and their current running command). Note: the username 'vokoun' is the surname/username of the ETE2AE course instructor. • uname -a: Prints comprehensive system info: operating system kernel name (Linux), network hostname (e.g., 'ete2ae.pef.czu.cz' - CZU is the Czech University of Life Sciences in Prague), kernel release version, build date, CPU architecture ('x86_64'), and operating system distribution.
[]	[]	Linux Command Pipelines <i>Explain how command pipelines work in Linux. Walk through the command chain used in OS Exercise 1.</i> Recall Keys: • Pipeline Principle: Uses the pipe symbol ' ' to pass the standard output (stdout) of one command directly as the standard input (stdin) of the next command. • Command Chain: <code>cat /home/vokoun/ete2ae grep '0dh26x;' grep 'Q65302rt39442408' cut -d ';' -f 6</code> 1. 'cat' reads and outputs the log file (stored in the directory of instructor 'vokoun'). 2. First 'grep' filters lines containing the session token '0dh26x'. 3. Second 'grep' filters lines containing the target question ID. 4. 'cut -d ';' -f 6' splits the output line by semicolon and extracts the 6th field (returning the correct answer '42').
[]	[]	SSH vs. Terminal Connection Security <i>Explain the primary security difference between SSH and traditional Telnet/Terminal connections.</i> Recall Keys: • SSH (Secure Shell): Connects via an encrypted channel (usually on Port 22), protecting all credentials and session data from interception. • Telnet/Terminal connection: Operates over unencrypted channels, transmitting usernames, passwords, and commands in plaintext (vulnerable to packet sniffing).
[]	[]	DNS Canonical Name (CNAME) & dig <i>What is a DNS CNAME record, and what command is used to find the true server name of www.pef.czu.cz? (OS Exercise 3)</i> Recall Keys: • CNAME (Canonical Name): A DNS record mapping an alias name to the true canonical domain name. • Command: Run <code>dig www.pef.czu.cz CNAME</code> (or <code>nslookup -type=cname</code>). This query reveals that the canonical server name hosting the site is <code>wlbnginx.czu.cz</code> (where czu.cz is Czech University of Life Sciences).
[]	[]	WHOIS Database Lookup <i>What is the function of the WHOIS utility, and what organization owns the IP 94.199.40.226? (OS Exercise 3)</i> Recall Keys: • Function: Queries registration databases to identify ownership details, registrant contacts, and allocated netblocks for domain names or IP addresses. • Organization: For '94.199.40.226', the WHOIS lookup returns Ministerstvo vnitra CR (which is Czech for Ministry of the Interior of the Czech Republic).
[]	[]	Ping Packet Size Customization <i>How do you customize the packet payload size in ping on Windows vs Linux, and why would you do this? (OS Exercise 3)</i> Recall Keys: • Commands: On Windows use <code>ping -l <bytes> <host></code>; on Linux use <code>ping -s <bytes> <host></code> (e.g. <code>ping -s 1400 www.seznam.cz</code> - where 'seznam.cz' is a popular Czech web portal). • Purpose: Tests latency and packet loss under custom packet sizes. Useful to test MTU (Maximum Transmission Unit) constraints, payload boundaries, and network packet fragmentation limits.

Step-by-Step Calculations

Master ed	Review	Question & Core Concepts Checklist
[]	[]	CPU Scheduling Walkthrough Walk through FCFS, SJF (non-preemptive), and Round Robin (Q=2) for: P1 (arrival 0, burst 6), P2 (arrival 1, burst 4), P3 (arrival 2, burst 2). Calculate Gantt charts and average waiting times. Recall Keys: • FCFS: - Gantt: P1 (0-6) → P2 (6-10) → P3 (10-12). - Waiting times: P1 = 0, P2 = 6-1 = 5, P3 = 10-2 = 8. - Avg Waiting Time: (0+5+8)/3 = 4.33 ms. • SJF: - At t=0, only P1 is ready (runs 0-6). - At t=6, both P2 (burst 4) and P3 (burst 2) are ready. SJF chooses P3 first. - Gantt: P1 (0-6) → P3 (6-8) → P2 (8-12). - Waiting times: P1 = 0, P3 = 6-2 = 4, P2 = 8-1 = 7. - Avg Waiting Time: (0+4+7)/3 = 3.67 ms (SJF is optimal!). • Round Robin (Q=2): - Queue transitions: t=0: [P1], t=2: [P2, P1(burst 4)], t=4: [P1(4), P3, P2(burst 2)]... - Gantt: P1 (0-2) → P2 (2-4) → P1 (4-6) → P3 (6-8) → P2 (8-10) → P1 (10-12). - Finish times: P3 = 8, P2 = 10, P1 = 12. - Turnaround (Finish - Arrival): P1 = 12, P2 = 9, P3 = 6. - Waiting (Turnaround - Burst): P1 = 12-6 = 6, P2 = 9-4 = 5, P3 = 6-2 = 4. - Avg Waiting Time: (6+5+4)/3 = 5.0 ms.
[]	[]	IP Subnetting Walkthrough For the host IP address 192.168.10.138/26, calculate the Subnet Mask, Network Address, Broadcast Address, First Usable IP, Last Usable IP, and Usable Host count. Recall Keys: • Subnet Mask: /26 means 26 network bits. In binary: `11111111.11111111.11111111.11000000` = 255.255.255.192. • Network Address: The block size is $2^{32-26} = 64$. Subnets rise in multiples of 64: .0, .64, .128, .192. The host IP .138 lies in the 192.168.10.128 subnet. • Broadcast Address: The next subnet starts at .192, so this subnet's broadcast is one address prior: 192.168.10.191. • First Usable Host: Network address + 1 = 192.168.10.129. • Last Usable Host: Broadcast address - 1 = 192.168.10.190. • Usable Hosts: $2^{32-26} - 2 = 2^6 - 2 = 64 - 2 = 62$ hosts.

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[]	[]	Page Replacement Walkthrough <i>Given reference string: 7, 0, 1, 2, 0, 3 with 3 memory frames. Walk through FIFO and LRU step-by-step and count page faults.</i> Recall Keys: • FIFO (First-In, First-Out): 1. Access 7: Frame [7, -, -] - Page Fault (1) 2. Access 0: Frame [7, 0, -] - Page Fault (2) 3. Access 1: Frame [7, 0, 1] - Page Fault (3) 4. Access 2: Replaces 7 (oldest) → Frame [2, 0, 1] - Page Fault (4) 5. Access 0: Already in frame → Frame [2, 0, 1] - Hit 6. Access 3: Replaces 0 (oldest) → Frame [2, 3, 1] - Page Fault (5) - Total Page Faults: 5 • LRU (Least Recently Used): 1. Access 7: Frame [7, -, -] - Page Fault (1) 2. Access 0: Frame [7, 0, -] - Page Fault (2) 3. Access 1: Frame [7, 0, 1] - Page Fault (3) 4. Access 2: Replaces 7 (least recently used) → Frame [2, 0, 1] - Page Fault (4) 5. Access 0: Already in frame (updates recency) → Frame [2, 0, 1] - Hit 6. Access 3: Replaces 1 (least recently used, since 0 and 2 were used more recently) → Frame [2, 0, 3] - Page Fault (5) - Total Page Faults: 5
[]	[]	802.1Q VLAN Frame Header Structure <i>Describe the structure and fields of an 802.1Q VLAN tag inserted into an Ethernet frame.</i> Recall Keys: An 802.1Q tag adds 4 bytes into the Ethernet frame header between the Source MAC and EtherType fields: 1. TPID (Tag Protocol Identifier): 2 bytes. Set to `0x8100` to identify it as an 802.1Q tagged frame. 2. TCI (Tag Control Information): 2 bytes, split into: • PCP (Priority Code Point): 3 bits. Used for Quality of Service (QoS) class of service. • DEI (Drop Eligible Indicator): 1 bit. Indicates if frame can be dropped during congestion. • VID (VLAN Identifier): 12 bits. Uniquely identifies the VLAN (values 1 to 4094; 0 and 4095 are reserved).

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[]	[]	IP Binary Conversion Walkthrough <i>Convert the decimal number 172 to binary, and the binary number 10011101 to decimal. Walk through the math.</i> Recall Keys: • Decimal to Binary (172): - Compare with powers of 2 (128, 64, 32, 16, 8, 4, 2, 1): - $172 \geq 128 \rightarrow$ bit is 1 ($172 - 128 = 44$) - $44 < 64 \rightarrow$ bit is 0 - $44 \geq 32 \rightarrow$ bit is 1 ($44 - 32 = 12$) - $12 < 16 \rightarrow$ bit is 0 - $12 \geq 8 \rightarrow$ bit is 1 ($12 - 8 = 4$) - $4 \geq 4 \rightarrow$ bit is 1 ($4 - 4 = 0$) - Remaining bits (2, 1) are 0. - Result: 10101100. • Binary to Decimal (10011101): - Multiply bits by their power-of-2 weights: - $1 \times 128 = 128$ - $0 \times 64 = 0$ - $0 \times 32 = 0$ - $1 \times 16 = 16$ - $1 \times 8 = 8$ - $1 \times 4 = 4$ - $0 \times 2 = 0$ - $1 \times 1 = 1$ - Sum: $128 + 0 + 0 + 16 + 8 + 4 + 0 + 1 = 157$.